

Operations Research & Mathematical Optimization

1st year Seminar 2026

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Why “operations research”?

Britain, before and during World War II

Scientists studied how to use radar effectively in military operations.

Research on **operations**: using scientific methods to improve decisions.

You have a goal, limited resources, and many possible plans.

How do you choose the best plan?

Source: INFORMS, “Useful history of operations research” (2015).

Example 1

Exploring the Moon



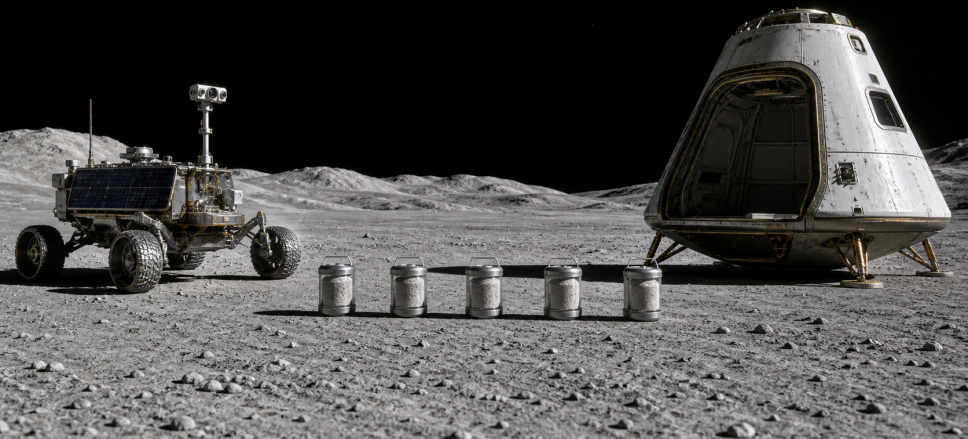
Apollo 11, 1969. Photo: NASA/JSC

We cannot bring everything home.

What should we bring back?

The last lunar return capsule

 One launch. Five samples. A 15 kg limit.



Five samples, one 15 kg limit



A. Ice-rich core

8 kg | 16
experiments

B. Volcanic rock

6 kg | 13
experiments

C. Deep-soil core

5 kg | 12
experiments

D. Meteorite

4 kg | 10
experiments

E. Glass beads

3 kg | 7
experiments

What should return to Earth?

	kg	experiments
☐ A. Ice-rich core	8	16
☐ B. Volcanic rock	6	13
☐ C. Deep-soil core	5	12
☐ D. Meteorite	4	10
☐ E. Glass beads	3	7

**One
launch**

**At most
15 kg**

Which samples
enable the most
experiments?

Two natural guesses tie

Take A and B

14 kg

29
experiments
enabled

or

Take C, D, E

12 kg

29
experiments
enabled

A less obvious combination is better

Take B, C, and D

Weight: $6 + 5 + 4 = 15$ kg

Experiments: $13 + 12 + 10 = 35$

$35 > 29$: these samples enable more science.

How do we find the best combination?

We found a selection worth **35 experiments**.

Could another selection be better?

How can we search **systematically**
and know when we are done?

Try every combination?

Each branch: **take** or **leave** a sample.

Start



Start with no decisions made.

Try every combination?

Each branch: **take** or **leave** a sample.

Start

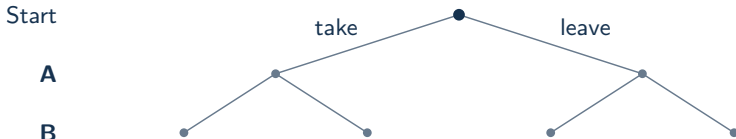
A



One sample. 2 possibilities.

Try every combination?

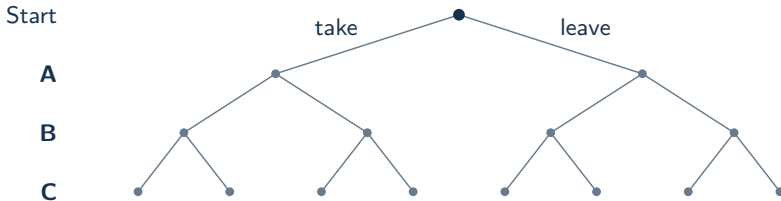
Each branch: **take** or **leave** a sample.



Two samples. 4 possibilities.

Try every combination?

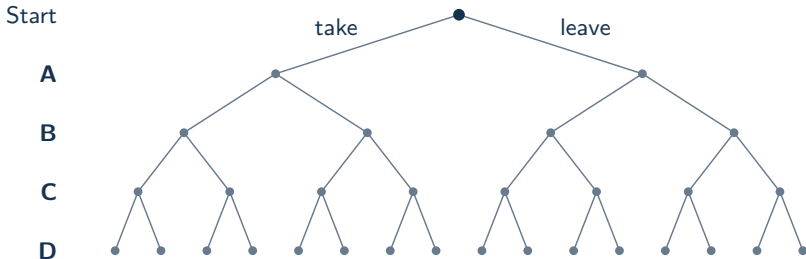
Each branch: **take** or **leave** a sample.



Three samples. 8 possibilities.

Try every combination?

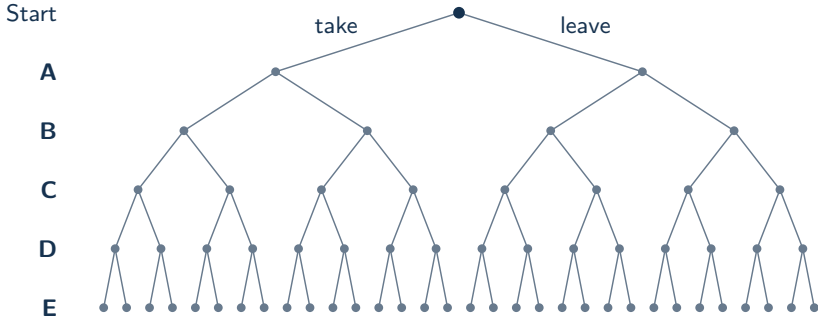
Each branch: **take** or **leave** a sample.



Four samples. 16 possibilities.

Try every combination?

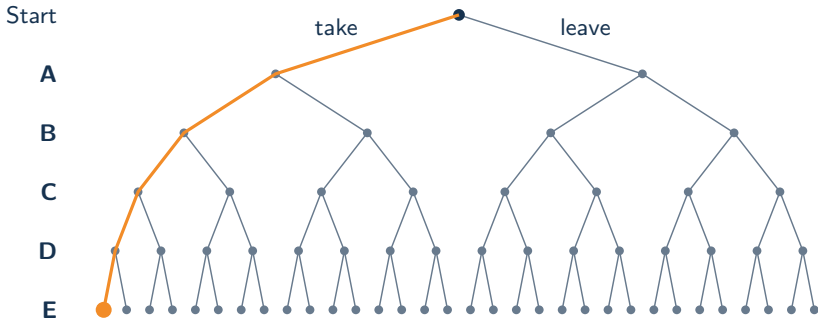
Each branch: **take** or **leave** a sample.



Five samples. 32 possible selections.

Check every selection

1 of 32 selections checked.



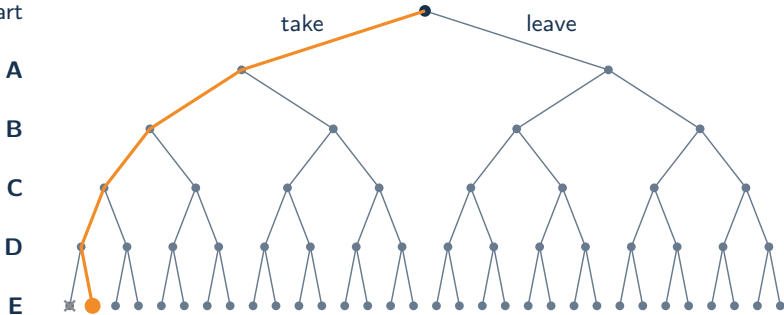
Current solution: **A B C D E** Experiments: 58 Weight: 26 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

2 of 32 selections checked.

Start



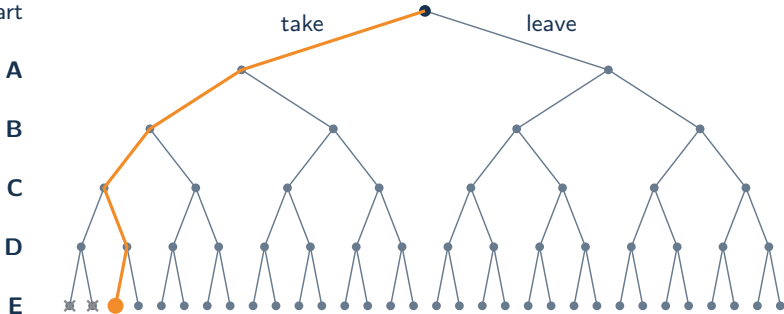
Current solution: **A B C D** E Experiments: 51 Weight: 23 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

3 of 32 selections checked.

Start

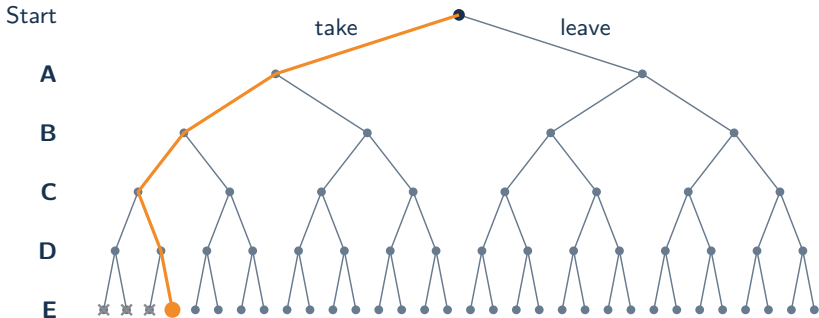


Current solution: **A B C D E** Experiments: 48 Weight: 22 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

4 of 32 selections checked.

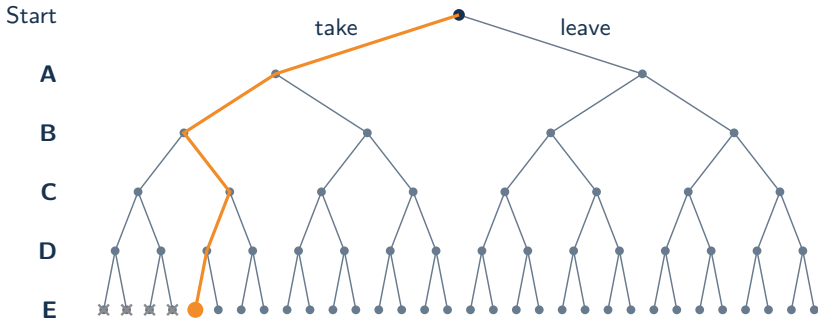


Current solution: **A B C** D E Experiments: 41 Weight: 19 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

5 of 32 selections checked.

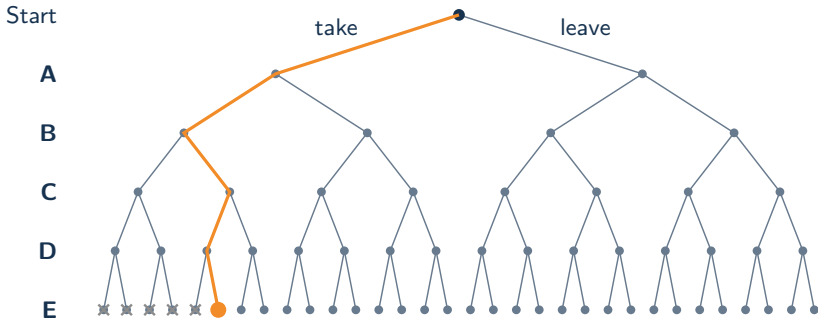


Current solution: **A B C D E** Experiments: 46 Weight: 21 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

6 of 32 selections checked.



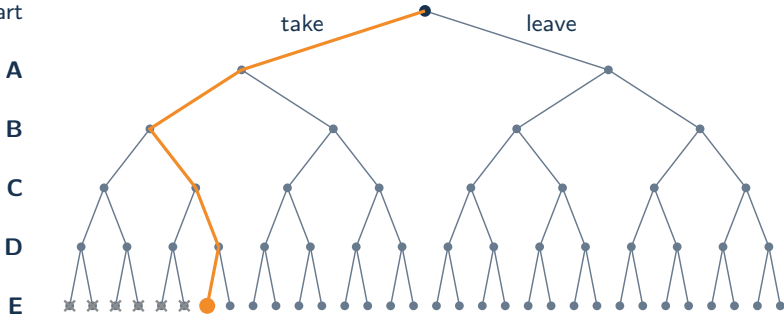
Current solution: **A B C D E** Experiments: 39 Weight: 18 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

7 of 32 selections checked.

Start

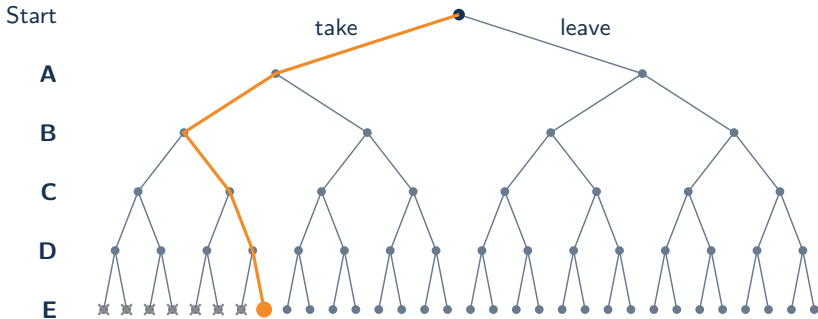


Current solution: **A B C D E** Experiments: 36 Weight: 17 kg (too heavy)

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

First feasible selection. Keep it?

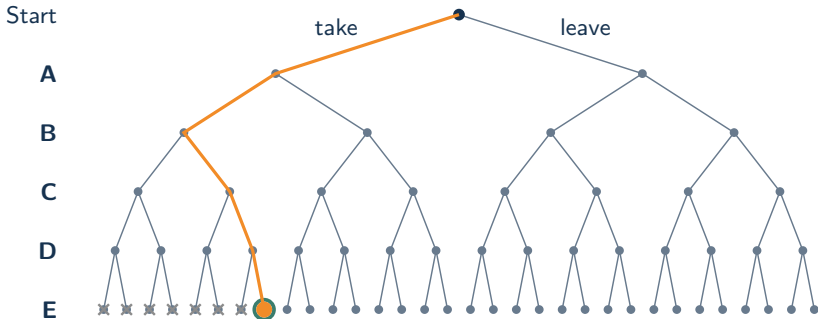


Current solution: **A B** C D E Experiments: 29 Weight: 14 kg

Best solution: A B C D E Experiments: – Weight: – kg none yet

Check every selection

Best solution updated: **29 experiments.**

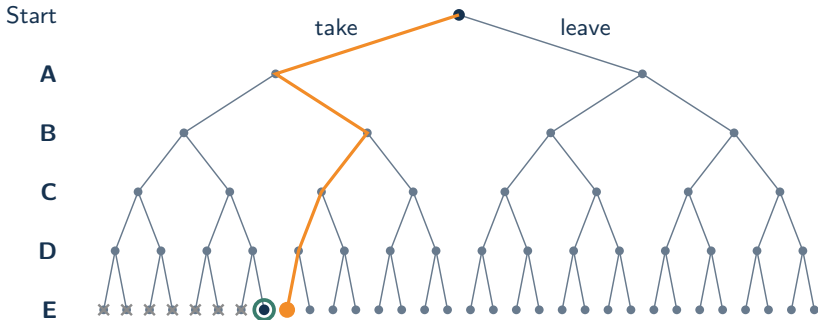


Current solution: **A B** C D E Experiments: 29 Weight: 14 kg

Best solution: **A B** C D E Experiments: 29 Weight: 14 kg

Check every selection

9 of 32 selections checked.

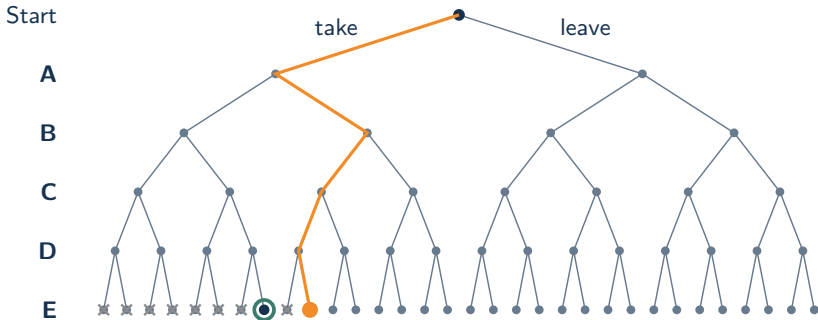


Current solution: **A** **B** **C** **D** **E** Experiments: 45 Weight: 20 kg (too heavy)

Best solution: **A** **B** **C** **D** **E** Experiments: 29 Weight: 14 kg

Check every selection

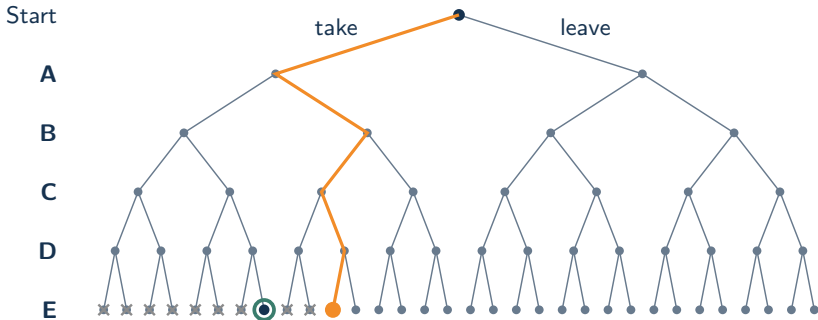
10 of 32 selections checked.



Current solution: **A** **B** **C** **D** E Experiments: 38 Weight: 17 kg (too heavy)

Best solution: **A** **B** C D E Experiments: 29 Weight: 14 kg

11 of 32 selections checked.



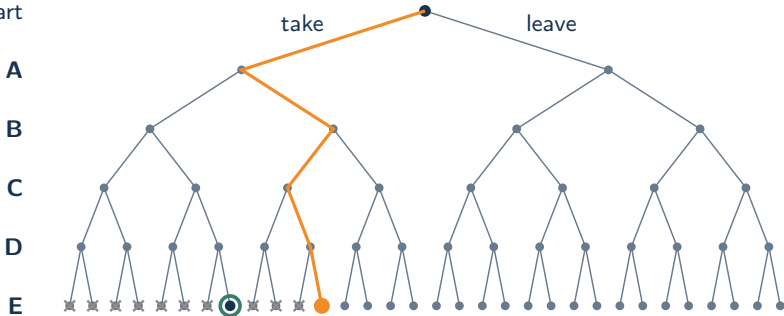
Current solution: **A** B **C** D **E** Experiments: 35 Weight: 16 kg (too heavy)

Best solution: **A B** C D E Experiments: 29 Weight: 14 kg

Check every selection

12 of 32 selections checked.

Start

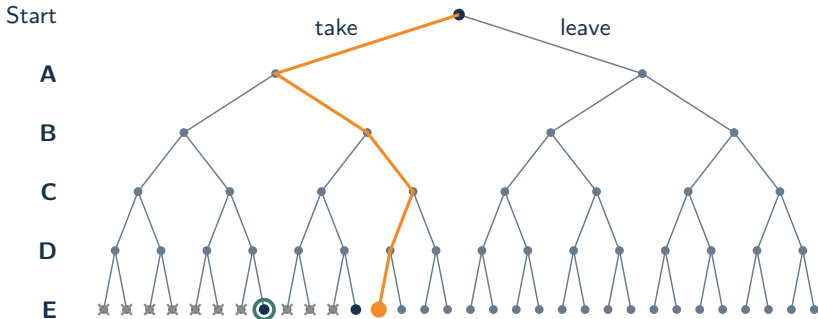


Current solution: **A** B **C** D E Experiments: 28 Weight: 13 kg

Best solution: **A** **B** C D E Experiments: 29 Weight: 14 kg

Check every selection

A better selection. Update the best?

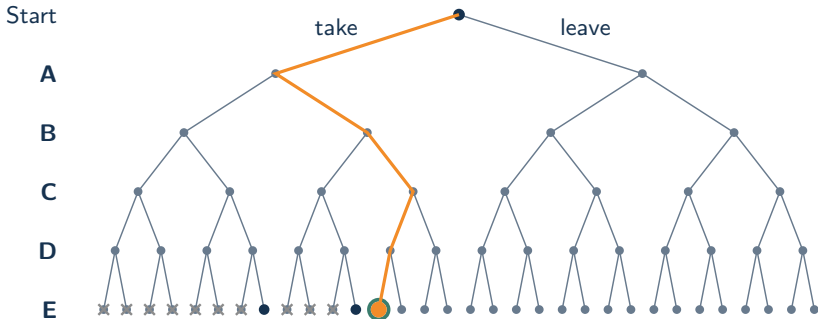


Current solution: **A** B C **D** E Experiments: 33 Weight: 15 kg

Best solution: **A** **B** C D E Experiments: 29 Weight: 14 kg

Check every selection

Best solution updated: **33 experiments.**



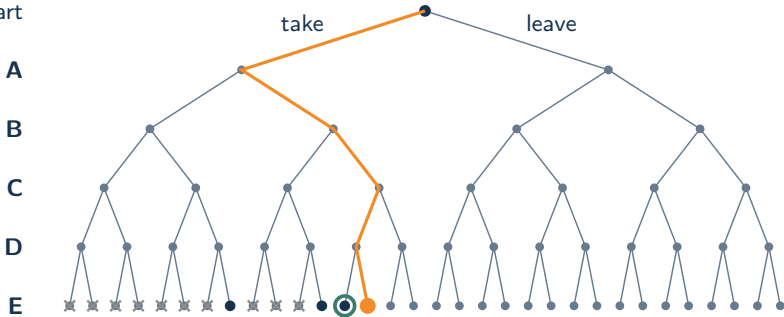
Current solution: **A B C D E** Experiments: 33 Weight: 15 kg

Best solution: **A B C D E** Experiments: 33 Weight: 15 kg

Check every selection

14 of 32 selections checked.

Start



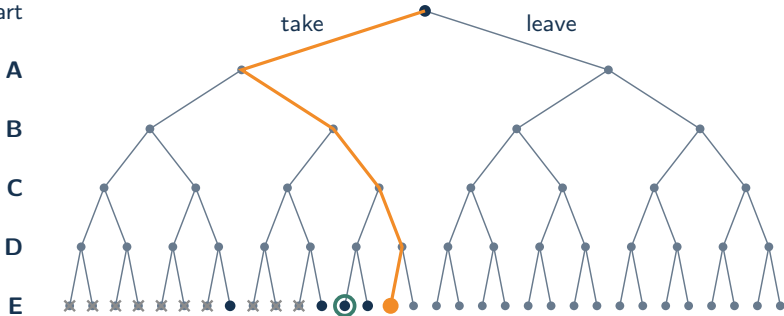
Current solution: **A** B C **D** E Experiments: 26 Weight: 12 kg

Best solution: **A** B C **D** E Experiments: 33 Weight: 15 kg

Check every selection

15 of 32 selections checked.

Start



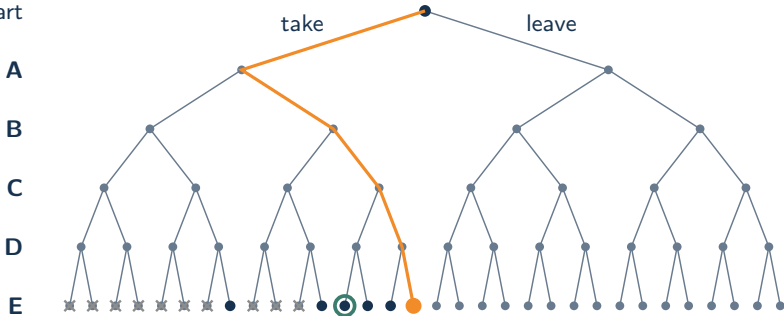
Current solution: **A** B C D **E** Experiments: 23 Weight: 11 kg

Best solution: **A** B C **D** **E** Experiments: 33 Weight: 15 kg

Check every selection

16 of 32 selections checked.

Start

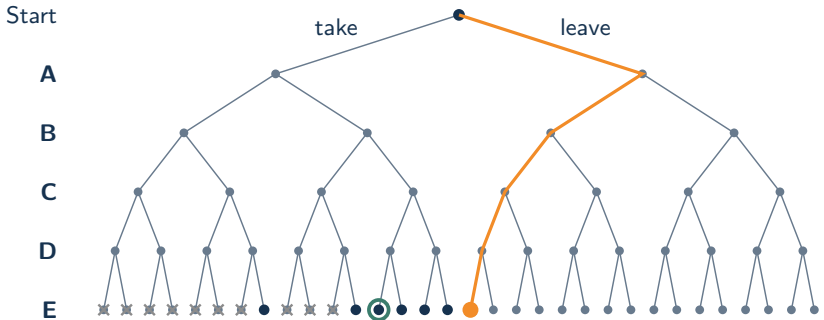


Current solution: **A** B C D E Experiments: 16 Weight: 8 kg

Best solution: **A** B C **D** **E** Experiments: 33 Weight: 15 kg

Check every selection

17 of 32 selections checked.

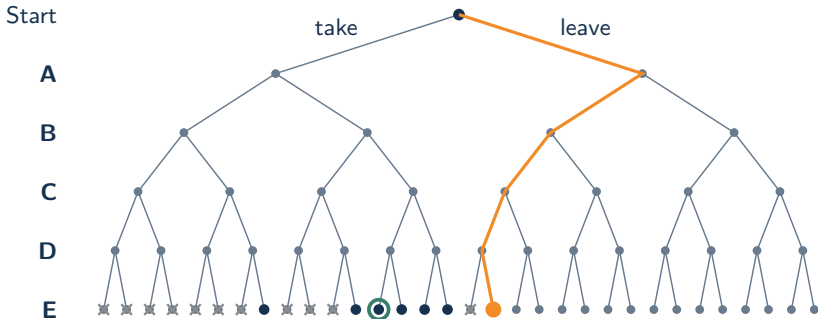


Current solution: A **B** **C** **D** **E** Experiments: 42 Weight: 18 kg (too heavy)

Best solution: **A** **B** **C** **D** **E** Experiments: 33 Weight: 15 kg

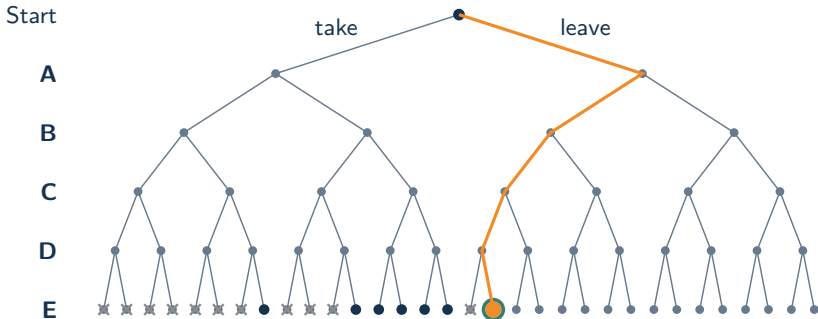
Check every selection

A better selection. Update the best?



Check every selection

Best solution updated: **35 experiments.**

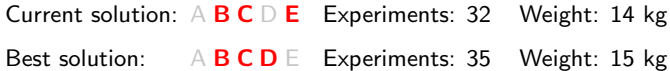


Current solution: A **B C D** E Experiments: 35 Weight: 15 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

19 of 32 selections checked.

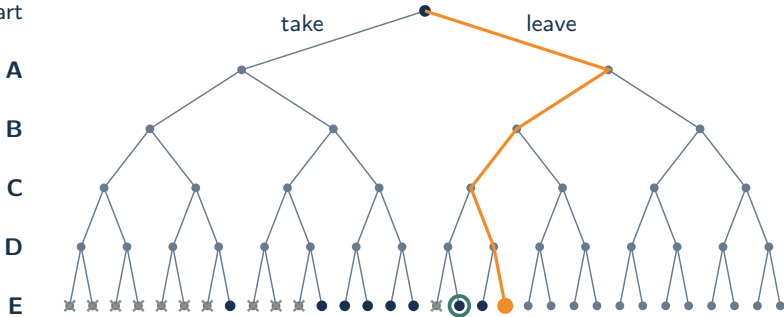
19 of 32 selections checked.



Check every selection

20 of 32 selections checked.

Start

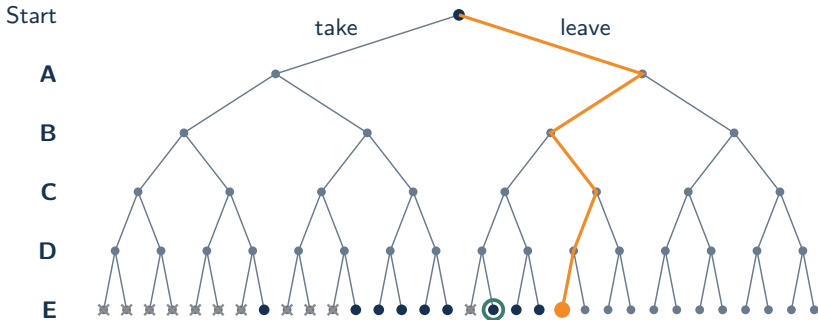


Current solution: A **B** **C** D E Experiments: 25 Weight: 11 kg

Best solution: A **B** **C** **D** E Experiments: 35 Weight: 15 kg

Check every selection

21 of 32 selections checked.

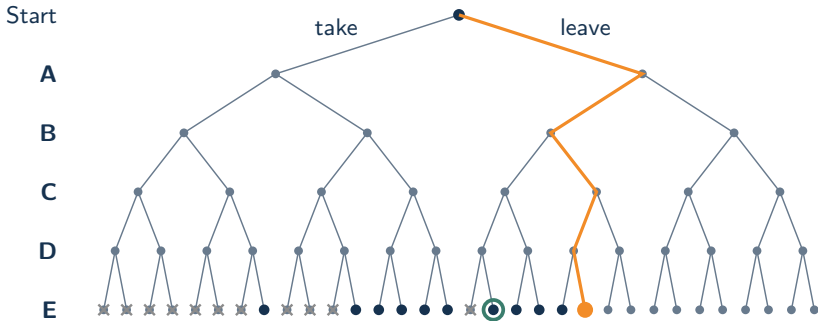


Current solution: A **B** C **D** E Experiments: 30 Weight: 13 kg

Best solution: A **B** **C** **D** E Experiments: 35 Weight: 15 kg

Check every selection

22 of 32 selections checked.



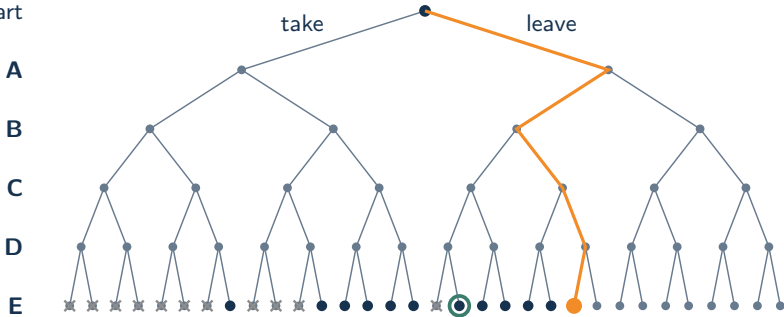
Current solution: A **B** **C** **D** E Experiments: 23 Weight: 10 kg

Best solution: A **B** **C** **D** E Experiments: 35 Weight: 15 kg

Check every selection

23 of 32 selections checked.

Start

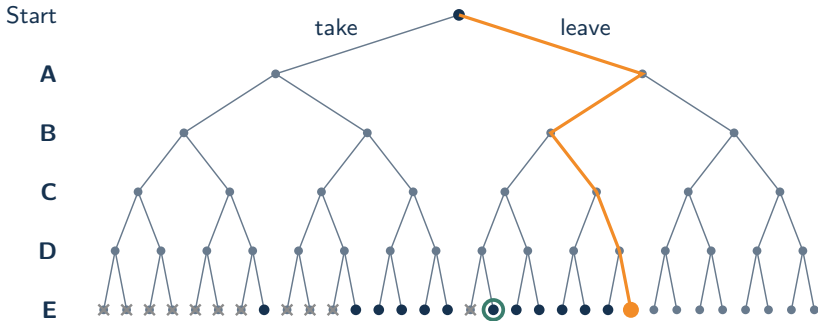


Current solution: A **B** C D **E** Experiments: 20 Weight: 9 kg

Best solution: A **B** **C** **D** E Experiments: 35 Weight: 15 kg

Check every selection

24 of 32 selections checked.

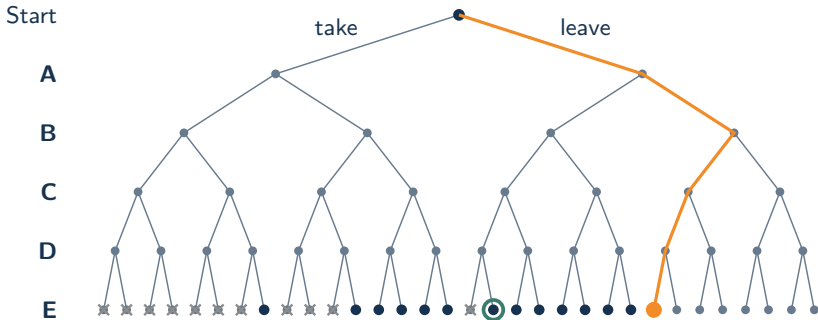


Current solution: A **B** C D E Experiments: 13 Weight: 6 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

Check every selection

25 of 32 selections checked.



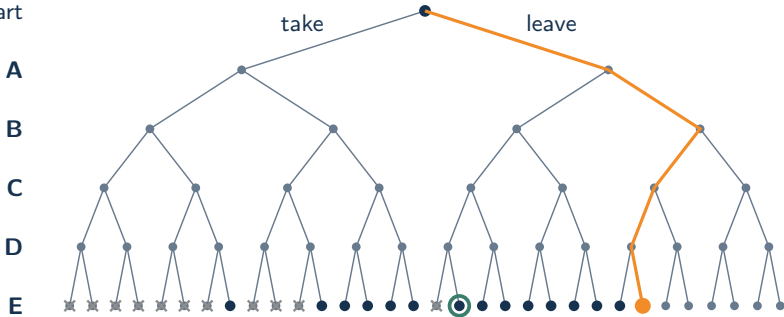
Current solution: A B **C D E** Experiments: 29 Weight: 12 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

Check every selection

26 of 32 selections checked.

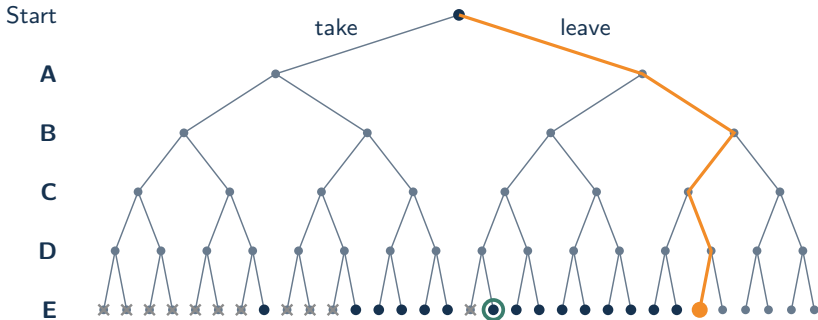
Start



Current solution: A B **C** **D** E Experiments: 22 Weight: 9 kg

Best solution: A **B** **C** **D** E Experiments: 35 Weight: 15 kg

27 of 32 selections checked.

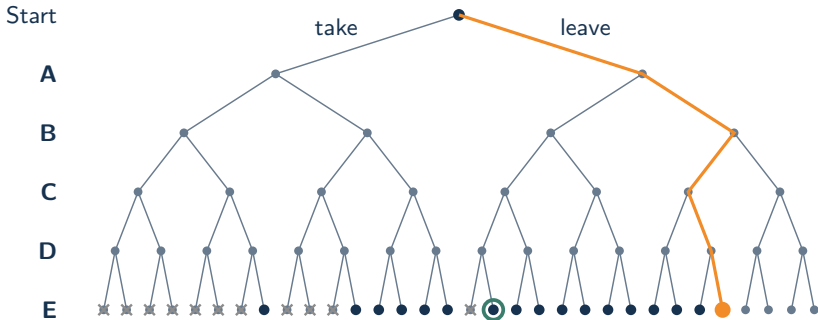


Current solution: A B **C** D E Experiments: 19 Weight: 8 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

Check every selection

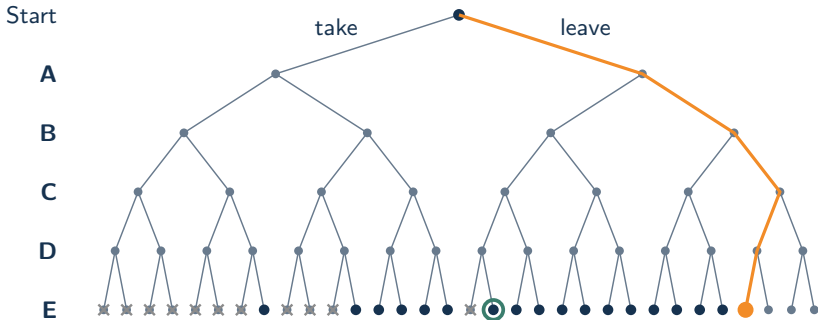
28 of 32 selections checked.



Current solution: A B **C** D E Experiments: 12 Weight: 5 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

29 of 32 selections checked.

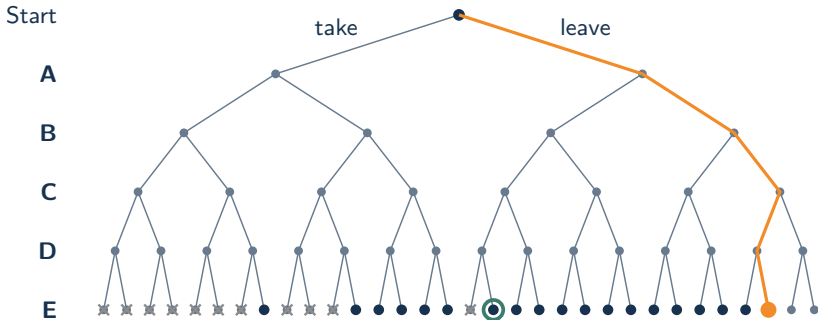


Current solution: A B C **D E** Experiments: 17 Weight: 7 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

Check every selection

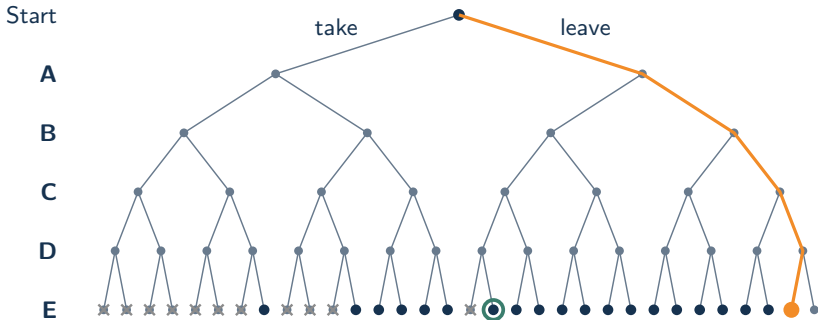
30 of 32 selections checked.



Current solution: A B C **D** E Experiments: 10 Weight: 4 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

31 of 32 selections checked.

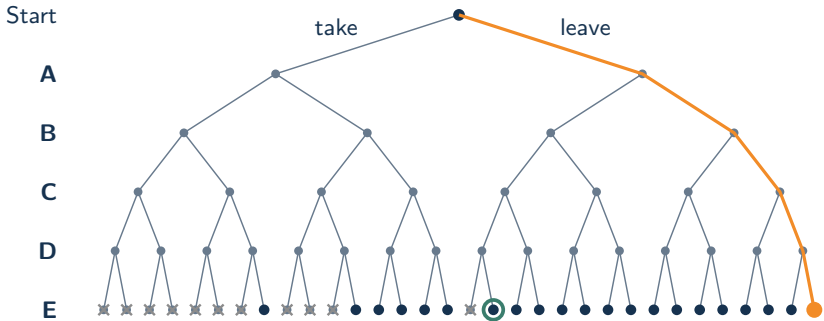


Current solution: A B C D **E** Experiments: 7 Weight: 3 kg

Best solution: A **B C D** E Experiments: 35 Weight: 15 kg

Check every selection

32 of 32 selections checked.

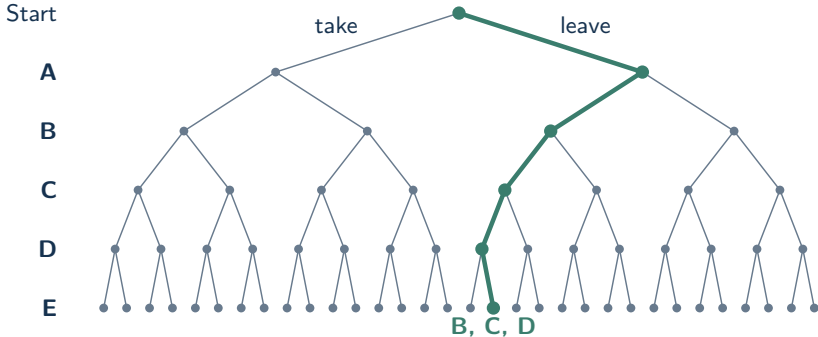


Current solution: A B C D E Experiments: 0 Weight: 0 kg

Best solution: A B C D E Experiments: 35 Weight: 15 kg

Check every selection

All 32 selections checked. **Best value: 35.**



Optimal selection: B, C, D.

Can we prove this without checking them all?

Combinatorial explosion

Every extra sample doubles the number of selections.

5 samples: $2^5 = 32$

30 samples: $2^{30} \approx 1$ billion

300 samples

$$2^{300} \approx 2 \times 10^{90}$$

possible selections



Observable universe

$$\sim 10^{80}$$

atoms (rough estimate)

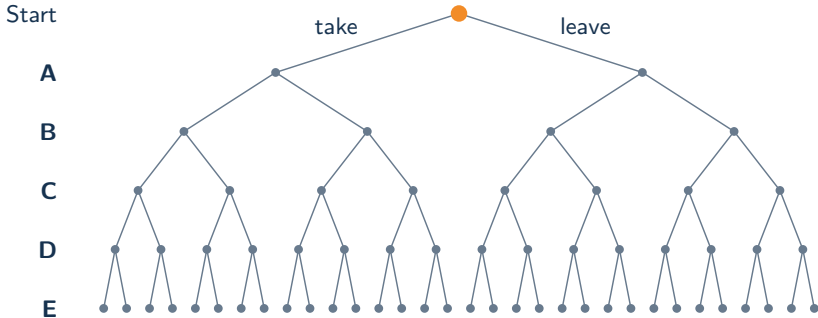
We need to skip possibilities, not just check them faster.

Selections counted before imposing the weight limit.

Source for the atom estimate: Peter Norvig, "On the (Small) Number of Atoms in the Universe" (2016).

Branch and bound: an illustration

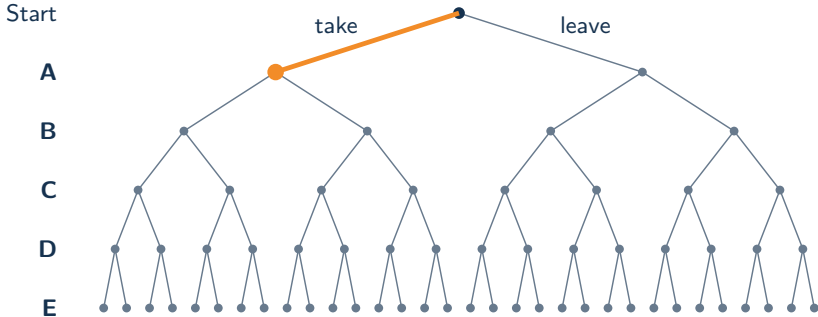
Start with a feasible solution.



Can we improve it without searching every branch?

Branch and bound: an illustration

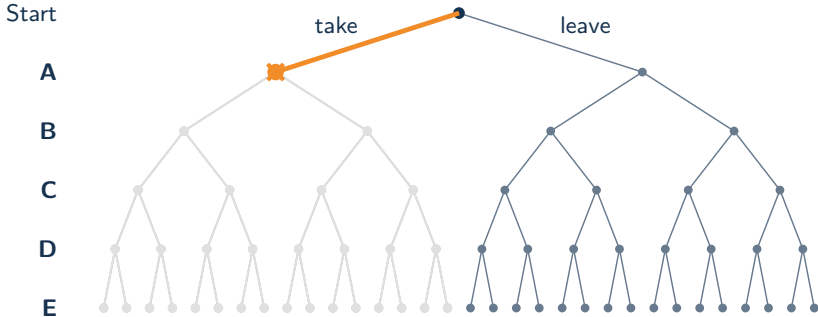
Explore a branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

Rule out this branch and everything below it.



One cut can eliminate many possibilities.

Branch and bound: an illustration

Backtrack.

Start

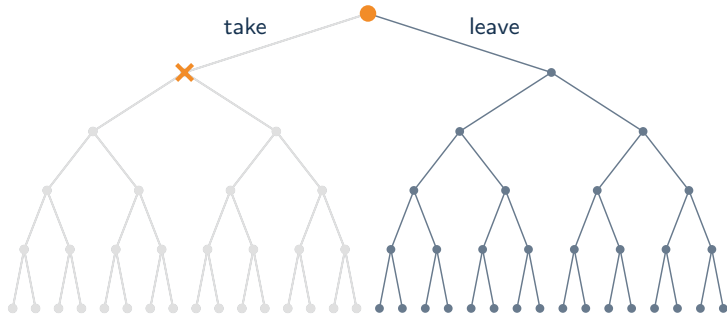
A

B

C

D

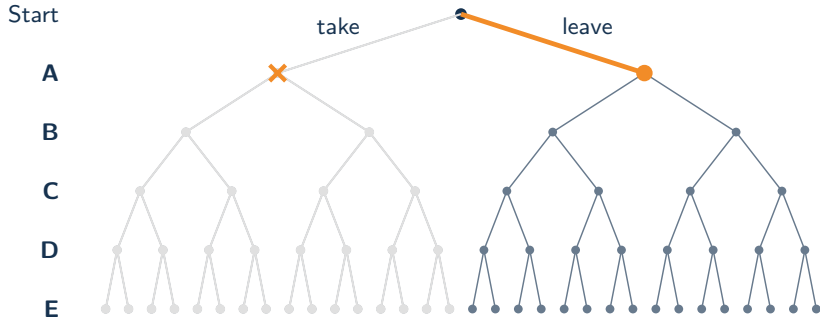
E



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

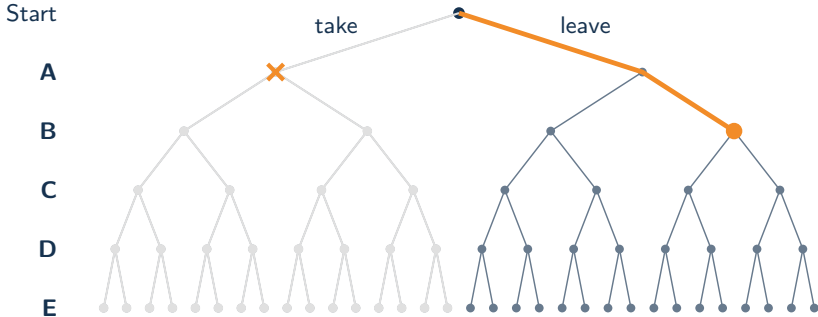
Try another branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

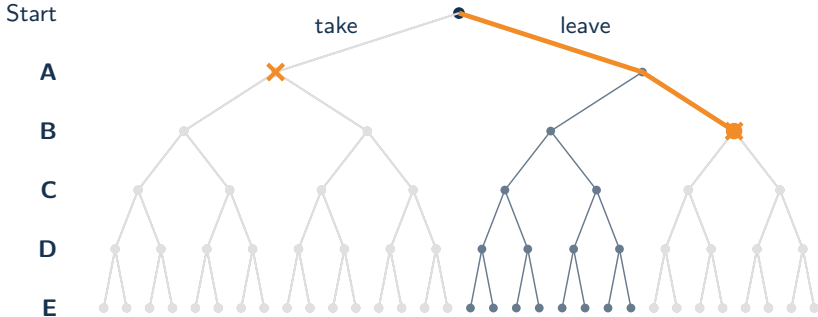
Explore a branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

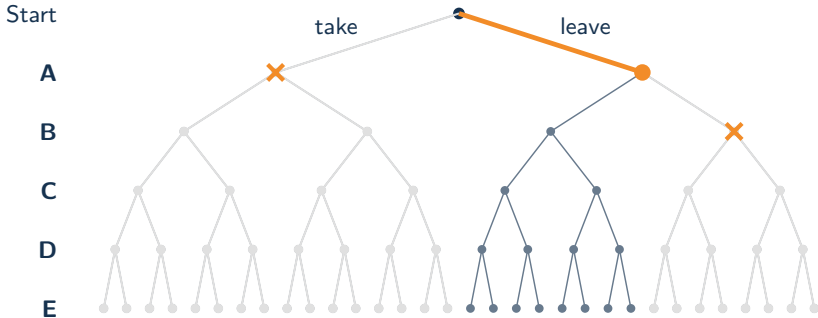
Rule out this branch and everything below it.



One cut can eliminate many possibilities.

Branch and bound: an illustration

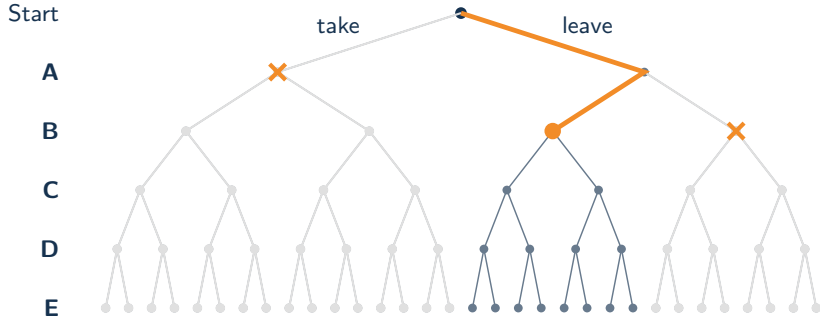
Backtrack.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

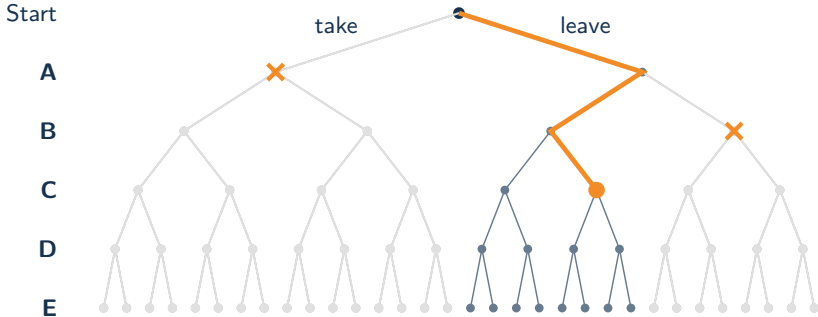
Try another branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

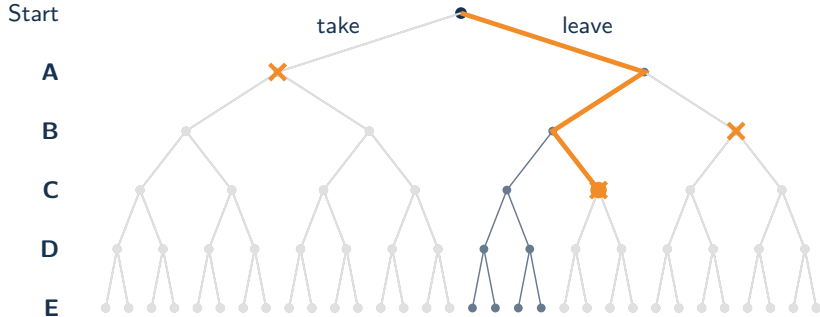
Explore a branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

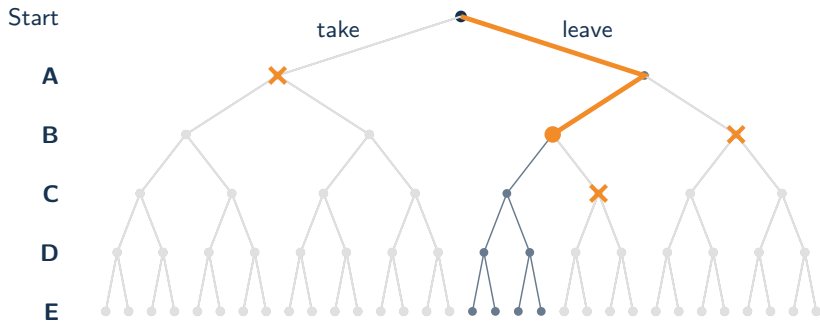
Skip this entire subtree.



One cut can eliminate many possibilities.

Branch and bound: an illustration

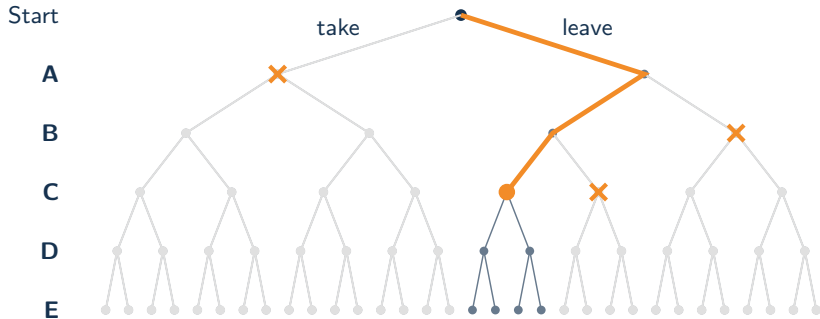
Backtrack.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

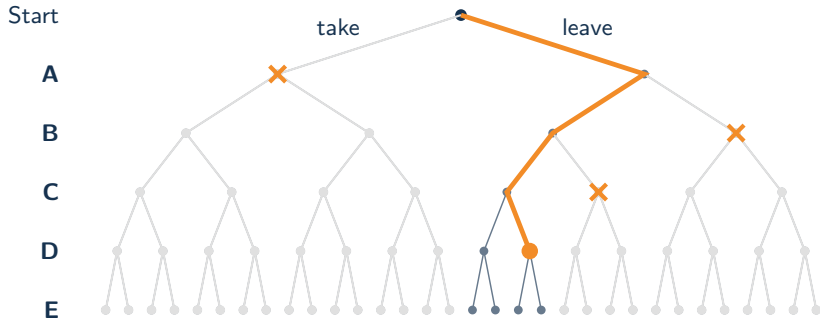
Try another branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

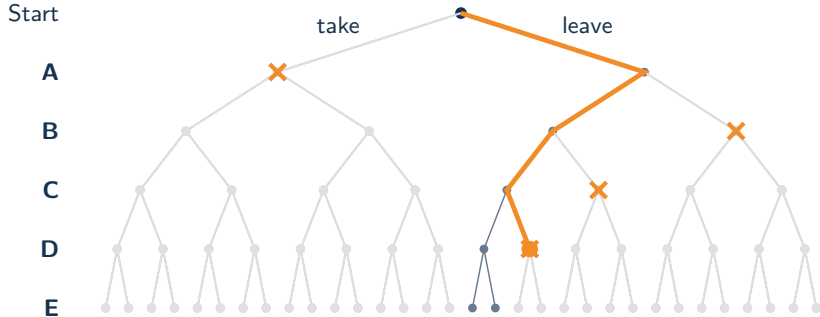
Explore a branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

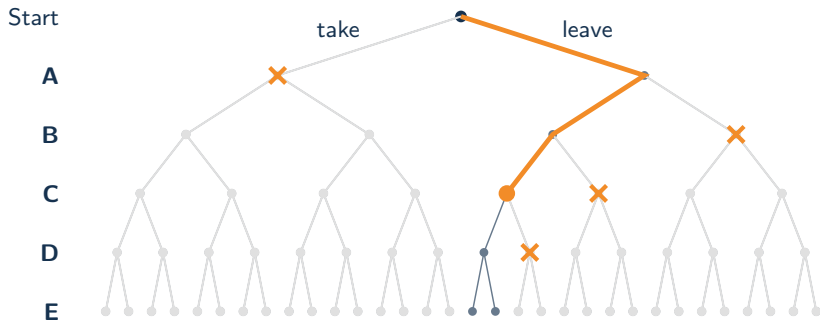
Skip this entire subtree.



One cut can eliminate many possibilities.

Branch and bound: an illustration

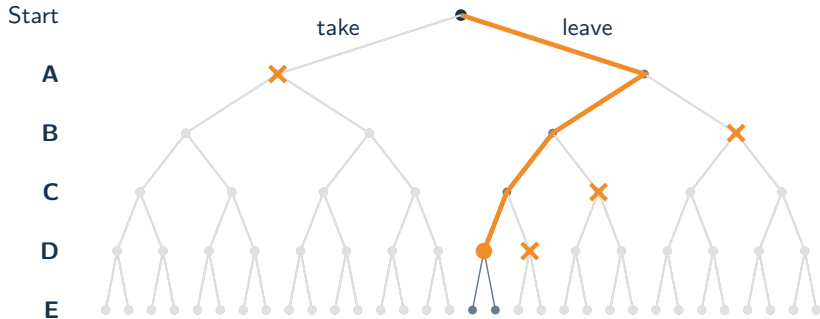
Backtrack.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

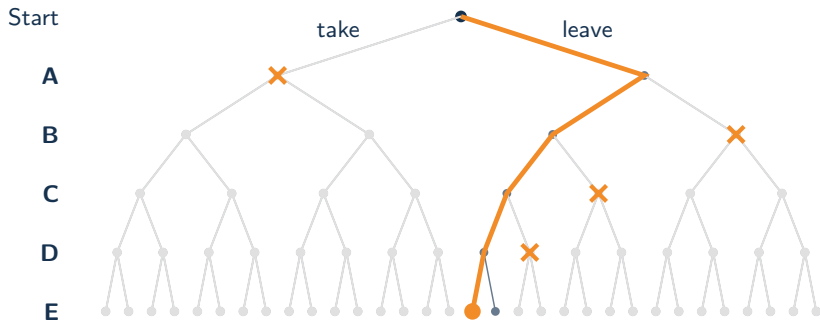
Try another branch.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

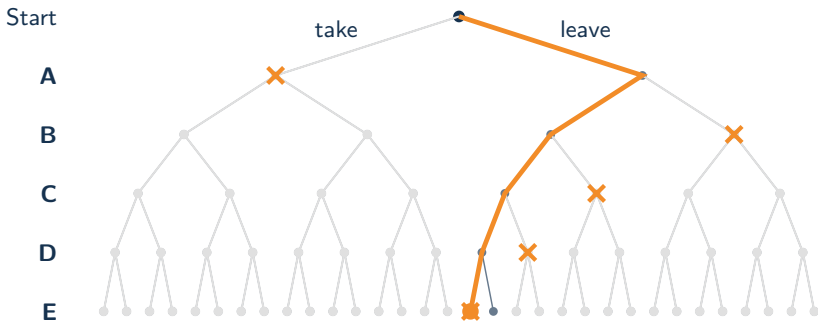
Check this selection.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

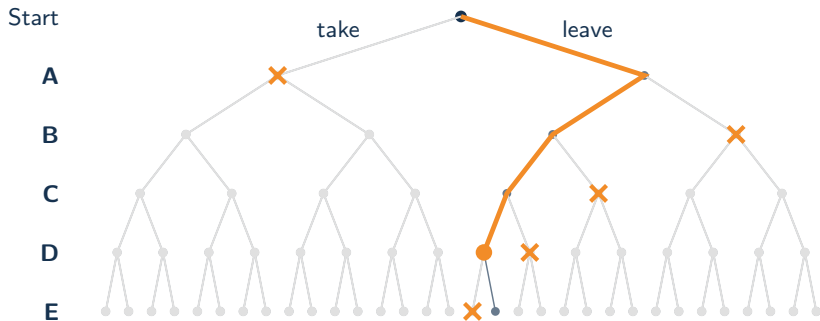
Rule out this selection.



One cut can eliminate many possibilities.

Branch and bound: an illustration

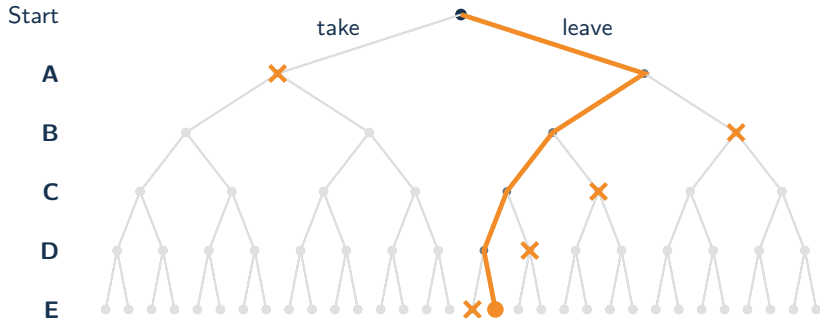
Backtrack.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

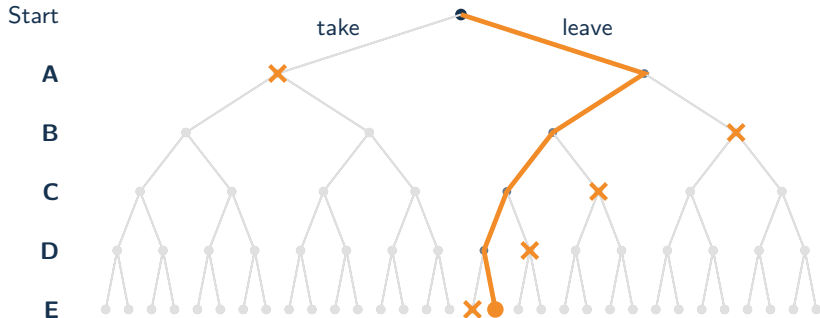
Check this selection.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

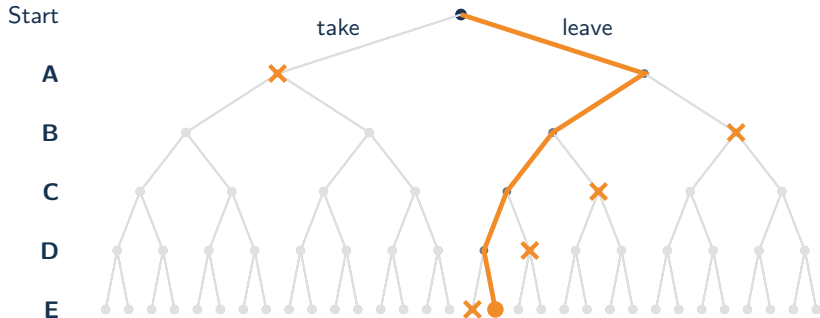
A better solution found.



Explore. Rule out branches. Backtrack.

Branch and bound: an illustration

Best solution confirmed.

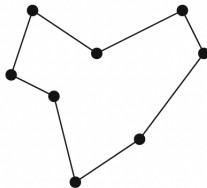


Much of the tree was never explored.

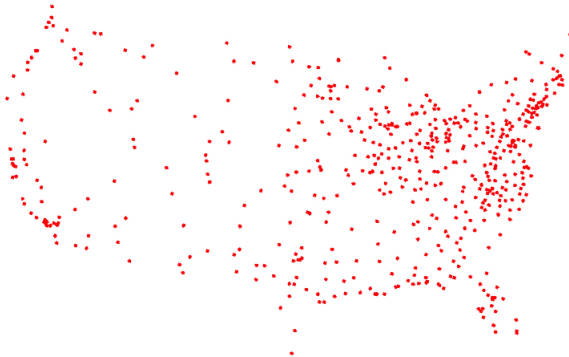
Example 2

The Traveling Salesman Problem

A traveling salesman must visit n cities and return to the city he started from. In which order should the salesman visit the cities to minimize the total cost of his tour?



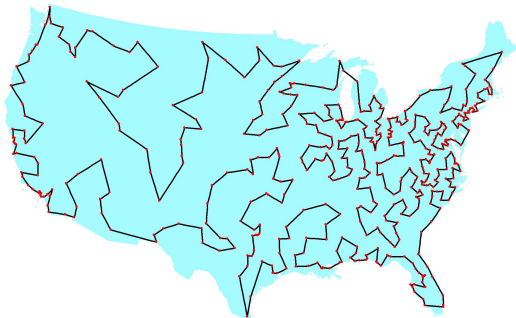
The Traveling Salesman Problem



The Traveling Salesman Problem

The data set *ATT532* consists of 532 cities located in the United States.

¹



¹https://www.math.uwaterloo.ca/tsp/history/tspinfo/att532_info.html

Routing decisions can save hundreds of millions

Who cares?

- ▶ UPS, FedEx, DHL
- ▶ Amazon Logistics
- ▶ Grocery and service fleets

**\$300–400
million**

projected annual cost
savings from UPS ORION
at full implementation

A shorter route means less fuel, less driver time, and more deliveries.

Source: UPS Investor Relations (2014 projection).

How many tours could we try?

Fix the starting city and order the remaining cities.

n cities: about $(n - 1)!$ tours

Just 70 cities

$\approx 1.7 \times 10^{98}$
possible tours



Observable universe

$\sim 10^{80}$
atoms (rough estimate)

At one billion tours per second, this would take about
 5.4×10^{69} trillion years.

More Examples

Applications in Machine Learning

The security of neural network is especially important for certain applications, e.g., self-driving cars. (Sun glare changes the picture slightly and unexpectedly.)



Which beams should treat the tumor?



Decision

Select the beam directions.

Requirement

Deliver enough dose to the tumor.

Goal

Protect surrounding healthy tissue.

Operations research

Identify important problems and
formulate them as mathematical models.

Mathematical optimization

Solve the mathematical models
to obtain good decisions.